A Systematization of Hallucination Detection in Large Language Models

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Abstract

Large language models can produce fluent and convincing outputs that are factually incorrect or unsupported, a problem commonly described as hallucination. Although this issue has been studied widely, prior work remains fragmented across tasks, benchmarks, and methodological assumptions, making it difficult to compare detection approaches in a unified way. This report presents a detection-focused Systematization of Knowledge on hallucination in large language models. Rather than organizing the field only by application domain, it develops a functional view of hallucination detection based on what different methods require, what they measure, and how they behave under different evidence conditions. The report introduces a unified taxonomy covering internal-signal, grounding-based, retrieval-based, verification-based, and tool-augmented approaches, and complements this analysis with a lightweight Streamlit prototype that supports side-by-side comparison of eight representative methods. A controlled 200-case benchmark spanning eight subject areas was used to evaluate the implemented methods under a shared interface. The results show that grounded and verification-oriented methods achieved the strongest and most balanced performance, while internal-signal methods were more conservative and less effective at detecting hallucinated content. The evaluation also shows that document-grounded memo cases are consistently harder than direct fact questions, highlighting the difficulty of evidence-conditioned judgment even in controlled settings. Overall, the report argues that hallucination detection is best understood as a comparative design space shaped by evidence assumptions, interpretability, and deployment trade-offs rather than as a single unified capability.

1 Introduction

Large language models (LLMs) have rapidly transformed artificial intelligence and natural language processing by changing how individuals and organizations access and utilize information [1], [2], [3], [4], [5]. Their strong performance across a wide range of language tasks has made them increasingly attractive for real-world deployment, but their growing influence has also exposed a major limitation: hallucination. In the context of LLMs, hallucination refers to outputs that are fluent, plausible, and often highly confident, yet factually incorrect or unsupported [6], [7], [8]. Hallucination is especially noticeable when LLMs are asked about topics beyond their knowledge or training exposure [6], [9]. Because the outputs can appear highly convincing and human-like [10], hallucinated content is often difficult to detect reliably in practice [2]. Hallucinations in large language models are generally categorized into two main types: intrinsic and extrinsic hallucinations. Intrinsic hallucinations occur when the generated output contradicts the source content, whereas extrinsic hallucinations introduce information that cannot be verified or supported by the source content [8], [11], [12]. While this intrinsic–extrinsic distinction remains widely used, more recent LLM-focused work also broadens the discussion to factuality and faithfulness hallucinations, reflecting the open-ended nature of modern LLM deployment [2].

The challenge becomes even more evident when these models are deployed in high-stakes domains such as healthcare, education, law, finance, and public information systems, intensifying concerns about factual reliability [9], [13], [14]. Hallucinated outputs can produce significant real-world harms, including misinformation, unsafe medical guidance, legal errors, and loss of public trust; in healthcare the consequences may be life-threatening [5], [13], [15].

Although hallucination has been widely studied, existing research remains fragmented [16]. Many studies address hallucination within specific natural language generation tasks, with insights often organized and evaluated using task-specific conventions. Valuable surveys exist in individual areas such as abstractive summarization [8], [12], while other works examine dialogue systems and more recent studies analyze hallucination in retrieval-augmented models or the theoretical limitations of probabilistic training objectives [9], [11], [16], [17]. Benchmarks such as TruthfulQA show that truthfulness does not improve reliably with scale alone, and that larger models can still generate persistent falsehoods [18]. As a result, cross-task comparability remains challenging because definitions, detection assumptions, and evaluation setups vary across settings. This fragmentation makes it difficult to directly compare hallucination detection methods across tasks, assess their generalization, or explain why they succeed in some contexts but fail in others.

This report addresses that gap through a detection-focused Systematization of Knowledge (SoK). Rather than surveying hallucination in a broad sense, it focuses specifically on how hallucinations are detected, how major detector families differ in the evidence they require, the assumptions they make, and the computational cost they impose, and how benchmark design can influence the appearance of progress [13]. Guided by this perspective, the report develops a unified view of hallucination detection and compares representative methods across summarization, question answering, dialogue, and retrieval-augmented generation, with particular attention to whether a detector remains useful across tasks and realistic deployment settings.

The main contributions of this work are four. First, it proposes a unified functional taxonomy of hallucination detection methods that organizes the literature by detector family and evidence assumptions rather than by narrow task boundaries. Second, it provides a cross-task comparative analysis of representative detector families, emphasizing transferability, interpretability, and deployment cost. Third, it critically examines the role of benchmarks and evaluation protocols in shaping claims of progress, including the influence of TruthfulQA [18] and retrieval-augmented generation evaluation settings [17], [19], [20]. Fourth, it presents a lightweight Streamlit prototype [21] that supports qualitative side-by-side inspection of representative detector families under a shared interface. Together, these contributions move beyond a conventional literature review by offering a more structured account of what current hallucination detection methods actually measure, where they perform well, where they fall short, and what future work is needed to make them more reliable in practice.

2 Related Work

Hallucination has been widely studied in both traditional natural language generation and modern large language models, and recent survey and overview papers provide broad coverage of its definitions, causes, detection strategies, mitigation methods, and evaluation practices [1], [2], [4], [5], [7], [16], [22], [23]. Earlier work in abstractive summarization played an especially important role in shaping the field. Maynez et al. [8] showed that fluent abstractive summaries can still be unfaithful to their source documents and formalized the influential distinction between intrinsic hallucinations, which distort source information, and extrinsic hallucinations, which introduce unsupported content. Their study further showed that hallucinations are common in summarization systems and that standard overlap metrics such as ROUGE and BERTScore correlate weakly with human judgments of faithfulness and factuality, motivating stronger semantic inference–based evaluation methods [8]. Ji et al. [16] note that hallucination in NLG is defined differently across tasks because tasks vary in their source grounding and tolerance for unsupported content. They further distinguish faithfulness to the input source from factuality with respect to world knowledge, warning against treating the two as interchangeable [16]. This earlier literature established the foundation for later LLM research by showing that linguistic quality and factual reliability are distinct objectives that must be evaluated separately.

As LLMs evolved into general-purpose systems rather than task-specific generators, the scope of hallucination research broadened substantially. Recent surveys argue that older task-bound definitions

such as intrinsic and extrinsic hallucination are no longer sufficient on their own because LLMs operate in open-ended settings where failures may involve not only contradiction with a provided source, but also conflict with real-world facts, user instructions, provided context, or internal logical consistency [2]. Huang et al. [2] therefore reorganize hallucination into factuality hallucination and faithfulness hallucination, with the former encompassing factual contradiction and factual fabrication, and the latter covering instruction inconsistency, context inconsistency, and logical inconsistency, while also tracing causes across data, training, and inference stages. Related survey work further frames hallucination as part of a broader trustworthiness and alignment problem rather than as an isolated generation error [23]. Liu et al. [23], for example, place hallucination within a seven-category trustworthiness taxonomy under the Reliability category, separate from the Misinformation sub-category, retain the intrinsic/extrinsic distinction, and review its possible causes, evaluation methods, case-study-based measurement, and mitigation strategies such as improved training-data quality, reward design in RLHF, and the use of external knowledge bases. Taken together, these surveys are highly valuable for mapping the field, but they remain largely panoramic. They provide breadth across definitions, causes, evaluation settings, and mitigation strategies, yet offer less direct guidance on how to compare hallucination detectors across tasks under a shared functional framework.

A second major strand of prior work focuses on benchmarks and datasets for evaluating hallucination. TruthfulQA is especially influential because it targets imitative falsehoods—cases where models generate false answers that mimic common human misconceptions [18]. Its results showed that large models can remain substantially less truthful than humans and that scaling alone does not reliably improve truthfulness [18]. However, TruthfulQA mainly probes open-ended question answering and adversarial truthfulness rather than source-grounded or retrieval-grounded failure modes. In response to such limitations, newer benchmarks have turned to retrieval-augmented settings. RAGTruth [17] is particularly important because it demonstrates that hallucinations persist even when supporting documents are provided; it contributes a large annotated corpus of nearly 18,000 naturally generated responses across question answering, data-to-text writing, and summarization tasks, with response-level and span-level annotations for RAG pipelines [17]. This progression from general truthfulness benchmarks to scenario-specific corpora marks an important advance, but it also deepens fragmentation because detectors are often evaluated under different task assumptions, annotation granularities, and evidence conditions. As a result, apparent improvements on one benchmark do not necessarily translate into robust performance across tasks or deployment settings.

A third major line of work examines inference-time methods for identifying and reducing hallucinations. One branch relies on internal model signals such as uncertainty or self-consistency. Sampling-based methods compare the semantic agreement of multiple generations, motivated by the idea that a model that truly knows an answer should respond consistently. Semantic Entropy Probes (SEPs) make this approach much cheaper in practice by using hidden states from a single generation to approximate semantic uncertainty, thereby avoiding the repeated sampling required by semantic entropy while still outperforming simpler probe-based baselines; however, the strongest multi-sample baselines can remain more accurate at substantially higher cost [22]. Another branch relies on explicit verification procedures. Chain-of-Verification (CoVe) is best understood as a self-verification method: the model first drafts an answer, then plans verification questions, answers them independently, and finally produces a revised response [24]. CRITIC extends this idea to tool-interactive verification by allowing LLMs to use external resources such as search engines, calculators, and code interpreters to critique and iteratively correct their outputs [25]. Taken together, these methods shift the focus from passive hallucination scoring toward active inference-time verification. At the same time, they blur the boundary between detection and mitigation, since they do not merely estimate hallucination risk but often use the verification signal to improve the final answer.

Prior work raises important doubts about whether language models can reliably diagnose or correct their own errors without external support. Huang et al. [3] show that intrinsic self correction is weak for reasoning tasks and often degrades performance when oracle labels or other external correctness signals are unavailable. CRITIC reaches a related conclusion from a different angle: its gains come from tool interactive verification, and the paper explicitly notes that exclusive reliance on self correction without external feedback yields only modest improvements or can even worsen performance [25]. These findings are important for hallucination detection because they challenge the assumption that the same model that produces an error can also be trusted to detect or repair it using only its internal capabilities. Beyond detection methodology, a growing body of theoretical and analytical work suggests that hallucination is not merely an engineering bug but is also tied to deeper

properties of language modeling. Hallucination is Inevitable [1] argues, within a formal framework where LLMs are treated as computable functions relative to computable ground truth functions, that hallucination cannot be completely eliminated for computable LLMs. Why Language Models Hallucinate [7] argues that current training and evaluation practices often reward guessing more than calibrated abstention, helping explain why plausible falsehoods persist. Related theoretical work on calibrated language models further shows that, for certain arbitrary facts, hallucination can arise from fundamental statistical tradeoffs even in an idealized setting, rather than from implementation flaws alone [9]. Together, these perspectives caution against overly optimistic interpretations of benchmark gains and highlight the need to examine carefully what different hallucination detectors actually measure.

Overall, the literature has made substantial progress in defining hallucination, building benchmarks, and proposing detection and verification strategies [2], [16], [25]. However, it remains fragmented across task settings such as summarization, open-domain question answering, dialogue, and retrieval-augmented generation [2], [16]. Existing surveys provide important breadth, but they typically organize the field around taxonomy, causes, metrics, mitigation strategies, or task-specific applications rather than around the operational assumptions of hallucination detectors themselves [2], [16]. Method papers such as CRITIC further highlight the practical importance of external-feedback-based verification, but they do not provide a detection-centered synthesis across methods [25]. For this reason, what is still missing is a detection-centered synthesis that compares methods by the evidence they require, the kinds of hallucinations they can expose, their transferability across tasks, and their computational practicality in realistic deployment settings. That gap motivates the present SoK.

3 Methodology

3.1 Overview

The prototype is implemented as a Streamlit application [21] that supports both single-method inspection and side-by-side comparison. In Quick Check mode, the user runs one detector on the current input. In Compare Methods mode, multiple detectors can be run on the same answer under a shared interface. The eight methods currently exposed are Internal-Signal Baseline, SEP-Inspired Internal Signal, Source-Grounded Consistency, Retrieval-Grounded Checker, RAG Grounded Check, Verification-Based Workflow, CoVe-Style Verification, and CRITIC-lite Tool Check. These methods are intentionally heterogeneous. Some are directly motivated by prior method families, including retrieval-augmented generation, Semantic Entropy Probes, Chain-of-Verification, and CRITIC [19], [22], [24], [25], while others are simplified project-defined detectors designed to operationalize broader internal-signal, source-grounded, and retrieval-grounded ideas under a shared interface. The prototype was designed this way to make comparisons across detector families more direct, interpretable, and analytically useful.

3.2 Shared Input Representation and Output Schema

All methods operate on a shared set of user-provided inputs, although each method uses only the fields relevant to its own assumptions. The available inputs include the original question or prompt, the answer to be evaluated, optionally sampled alternative answers, an optional source passage, an optional evidence block, and optionally uploaded documents. The interface presents the inputs relevant to the current method selection. In Quick Check mode, it shows only the fields used by the selected method. In Compare Methods mode, it shows the combined set of inputs used across the selected methods, while each detector still operates only on the fields relevant to its own design. This structure was chosen so that different detector families could be compared within a shared interface without forcing every method to rely on the same evidence conditions.

Method-specific processing begins inside each detector, but several methods share common text-processing utilities. In particular, grounded, retrieval-based, and verification-oriented methods normalize relevant text inputs and extract basic answer-level claims from the submitted response. These claims then serve as a common working unit for grounded checking, retrieval-backed checking, and staged verification workflows. The system returns results in a shared detector schema so that different methods remain comparable. Each result includes the method name, detector family, normalized hallucination-risk score, qualitative risk label, confidence estimate, explanation, evidence records, citations, intermediate steps, optional revised answer, latency information, and method-

specific metadata or details. This schema supports direct comparison across methods, since the prototype is aimed at comparative analysis rather than isolated single-method evaluation.

Risk is represented on a normalized scale and then mapped to low, medium, or high risk for consistency in the interface. For methods that assess answers at the claim level, individual claim judgments are combined into an overall score. When a clear contradiction is identified, it is weighted more heavily than a simple lack of support so that stronger errors have a greater effect on the final risk estimate [16].

3.3 Shared Grounding and Retrieval Pipeline

Grounded and retrieval-based methods use a shared local evidence pipeline. The system first collects any available source passage, evidence text, and uploaded documents into a local evidence set. These materials are then divided into short chunks, typically at the sentence level or over small text windows, so that retrieval results and citations remain easier to interpret. Each chunk is kept together with document identifiers, source labels, and related metadata so that supporting evidence can be traced later in the analysis.

A local vector index is built over these chunks. When a dense embedding backend is available, the system uses sentence-transformer embeddings; otherwise, it falls back to a TF-IDF representation so that retrieval can still operate in lightweight environments. For each claim, the system compares the claim with the indexed chunks and retrieves the most relevant ones. These retrieved chunks are then evaluated using a combination of semantic similarity, lexical overlap, token coverage, phrase overlap, entity overlap, numeric alignment, and explicit contradiction cues. Based on this analysis, each claim is assigned a status such as supported, weakly supported, contradicted, or insufficiently grounded.

The same retrieval process is reused across several methods, but not always in the same role. For some methods, it serves as the main way of checking whether an answer is hallucinated. For others, it mainly provides supporting evidence for a larger verification or critique step. Using the same retrieval process across methods also makes the comparison more fair, since differences in the results are more likely to come from the method itself rather than from completely different evidence pipelines.

3.4 Internal-Signal Methods

The first detector family relies on internal model signals rather than external evidence [22]. The Internal-Signal Baseline estimates hallucination risk from uncertainty-related signals for a single answer. When the local Hugging Face backend is available, the method scores the answer with a local causal language model and extracts features such as negative log probability, variation in token-level confidence, entropy-based summaries, and hidden-state statistics. These signals are then combined into a risk estimate that reflects how stable or uncertain the answer appears internally. When the required backend is not available, the method falls back to a deterministic text-based proxy that uses surface characteristics of the answer as an approximate uncertainty signal. This fallback is treated as a lightweight approximation rather than a full internal probability-based estimate.

The SEP-Inspired Internal Signal method builds on the same general idea, but adds hidden-state features and, when available, information from sampled alternative answers. In this prototype, it is better understood as a lightweight SEP-inspired method than as a full implementation of Semantic Entropy Probes. It uses hidden-state statistics from one or more answers and can also use sampled alternatives to look at how consistent or variable the responses are across generations. If an offline probe model is available, it is applied to the extracted feature vector; otherwise, the method falls back to a local feature-based approximation. This allows the method to compare single-answer uncertainty with patterns observed across multiple answers, while making clear that it remains an approximation rather than a full reproduction of the original SEP approach.

3.5 Grounding-Based Methods

The second family evaluates answers against provided textual evidence [19], [20]. Source-Grounded Consistency is the simplest method in this group. It assumes that a reference passage is already available and checks whether the extracted answer claims are supported by that source [8]. The source is divided into smaller snippets, and each claim is compared against them using local similarity signals. Based on that comparison, each claim is labeled as supported, weakly supported, contradicted, unclear,

or unsupported. This method is relatively simple and easy to interpret, and it serves as a transparent baseline for source-grounded tasks such as summarization or grounded question answering.

Retrieval-Grounded Checker extends this idea to cases where supporting evidence must be retrieved from a larger local evidence set [19], [20]. Instead of assuming a single source passage, it builds an index over the available evidence text and uploaded documents, retrieves the most relevant chunks for each claim, and evaluates whether those chunks provide support, contradiction, or too little evidence to decide. The method also returns citations and claim-level evidence traces, which makes its decisions easier to inspect in the current system. Because it checks claims against a broader indexed evidence set rather than a single provided passage, it is more flexible than Source-Grounded Consistency while still remaining fairly interpretable.

RAG Grounded Check uses the same retrieval pipeline, but applies it in a retrieval-augmented generation setting [19], [20]. In this method, the system first retrieves relevant evidence and then checks whether the answer is actually grounded in that retrieved material [17], [19], [20]. It is still a post hoc grounded checker over a retrieved evidence set rather than a full end-to-end RAG generation pipeline [19], [20]. The method distinguishes between claims that are contradicted by the retrieved evidence, claims that remain unsupported even after retrieval, and cases where the retrieved evidence is not strong enough to support a clear judgment [17]. This makes it useful for separating retrieval failure from answer-level hallucination.

3.6 Verification and Tool-Augmented Methods

The third family shifts from passive scoring to active verification [24], [25]. Verification-Based Workflow is the simplest method in this group. It breaks the answer into claims, turns those claims into verification questions, and checks them against local evidence. Each claim is then marked as verified, contradicted, or unresolved, and the method returns a summary of that process. Compared with the more structured CoVe-style and CRITIC-style pipelines, this method is simpler and mainly serves as a basic reference point for staged verification.

CoVe-Style Verification follows a verify-and-revise process inspired by Chain-of-Verification [24]. In this method, the initial answer is broken into claims, which are then turned into verification questions and checked against retrieved evidence. The supported findings are used to produce a revised answer. In the prototype, this revision step is deterministic and evidence-based rather than driven by a large multi-step LLM pipeline. It was designed this way to keep the method relatively lightweight while still preserving the main CoVe idea of separating drafting, verification, and revision rather than relying on a single-pass judgment.

CRITIC-lite Tool Check extends staged verification by using local tools to evaluate individual claims [25]. In the current implementation, the main tools are local retrieval and numeric consistency checking. Claims are checked with the relevant tools, after which the answer is critiqued based on the tool outputs and, when needed, revised into a more grounded version. If the answer changes substantially, the system can optionally run a second pass to check the revision again. The method is called CRITIC-lite because it captures the general idea of tool-supported critique and revision without attempting to reproduce the full scope of CRITIC, which uses a broader range of external tools such as web search, calculators, and code interpreters [25].

3.7 Comparative Scoring Logic

To support comparison across detector families, the prototype uses a shared scoring logic across different methods. Each method produces either a direct answer-level score or a set of claim-level judgments that are combined into a single final score. Supported evidence is mapped to low risk, unresolved or weakly supported evidence to medium risk, and contradictions or strongly unsupported claims to high risk. In grounded and verification-based methods, a clear contradiction is given more weight so that it can drive the final score even when other parts of the answer appear acceptable.

The prototype also keeps some of the intermediate information produced by each method. For retrieval-based methods, this usually means the retrieved chunks, their scores, and the evidence that was selected. For verification-based methods, it includes the verification questions, the answers supported by evidence, the critique steps, and any revised answer. For internal-signal methods, it

mainly includes feature summaries and backend status information. Keeping this information makes it easier to understand how each method reached its final judgment.

3.8 Scope and Implementation Honesty

An important design choice in this project was to keep the scope controlled. The prototype was built as a qualitative comparison framework and as a companion demo for the SoK, rather than as a production system or a full benchmarking platform. For that reason, some methods were implemented in greater detail, while others were included in lighter form as approximations or comparison baselines. In the current prototype, Internal-Signal Baseline, Retrieval-Grounded Checker, CoVe-Style Verification, and CRITIC-lite Tool Check are the methods with the most complete implementations, while SEP-Inspired Internal Signal, RAG Grounded Check, Source-Grounded Consistency, and Verification-Based Workflow are included in lighter form for comparison. Some demonstration questions and prototype test cases were generated with AI assistance and then reviewed and refined manually for clarity, balance, and relevance to the detector behaviors being examined. The goal was to compare detector families within a shared framework while remaining clear about differences in implementation detail, evidence assumptions, and practical cost.

4 Experiments and Results

4.1 Evaluation Design

The experiments were designed as a controlled comparison of the implemented methods rather than as a broad claim about hallucination detection in general. The goal was to examine whether the methods showed clear and practically meaningful differences under a shared interface, while keeping the benchmark challenging enough to be useful without making it so adversarial that all methods performed poorly.

A 200-case benchmark was created and run on the current codebase. The final set included 25 cases in each of eight subject areas: Geography, Science, Culture, Health, Public Policy, Environment, Education, and Technology. At the overall dataset level, the benchmark was balanced between 100 supported low-risk cases and 100 hallucinated high-risk cases. It combined 130 direct fact questions with 70 document-grounded memos so that the evaluation covered both short factual checks and evidence-based judgments.

This setup was intended to support comparison across detector families while still preserving some variation in task structure. Direct fact questions tested short-form factual checking, whereas memo cases introduced explicit evidence conditions that made it easier to compare grounded, retrieval-based, and staged verification methods.

Each method was evaluated on the same 200 cases. For quantitative scoring, a supported case was counted as correct only when the method returned a Low risk label, while a hallucinated case was counted as correctly detected when the method returned either Medium or High risk. Precision, recall, and F1 therefore correspond to the hallucinated, or elevated-risk, class. Because the methods used slightly different internal score ranges, the reported average risk scores were normalized to the [0, 1] interval before aggregation.

4.2 Overall Quantitative Results

Overall performance fell within a fairly consistent range. Accuracy varied from 83.0% to 90.5%, with the highest result coming from Source-Grounded Consistency at 90.5%. Retrieval-Grounded Checker, RAG Grounded Check, Verification-Based Workflow, CoVe-Style Verification, and CRITIC-lite Tool Check each reached 90.0%. SEP-Inspired Internal Signal reached 88.0%, while Internal-Signal Baseline reached 83.0%. Overall, the internal-signal methods performed reasonably well, but they remained weaker than the grounded and verification-based methods in this benchmark.The overall quantitative results are summarized in Table 1 and figure 1 provides a visual comparison of overall accuracy across methods.

The precision and recall results show a clear difference between the detector families. Internal-Signal Baseline was highly conservative. It achieved 98.5% precision but only 67.0% recall, which means that it rarely flagged supported answers incorrectly but missed many hallucinated ones. SEP-Inspired

Table 1: Overall results across the 200-case benchmark.

Method	Accuracy (%)	Precision (%)	Recall (%)	F1 (%)	Avg. score on supported cases	Avg. score on hallucinated cases
Internal-Signal Baseline	83.0	98.5	67.0	79.8	0.008	0.469
SEP-Inspired Internal Signal	88.0	100.0	76.0	86.4	0.000	0.676
Source-Grounded Consistency	90.5	89.3	92.0	90.6	0.179	0.897
Retrieval-Grounded Checker	90.0	89.2	91.0	90.1	0.194	0.866
RAG Grounded Check	90.0	89.2	91.0	90.1	0.197	0.894
Verification-Based Workflow	90.0	89.2	91.0	90.1	0.195	0.878
CoVe-Style Verification	90.0	89.2	91.0	90.1	0.194	0.866
CRITIC-lite Tool Check	90.0	89.2	91.0	90.1	0.230	0.879

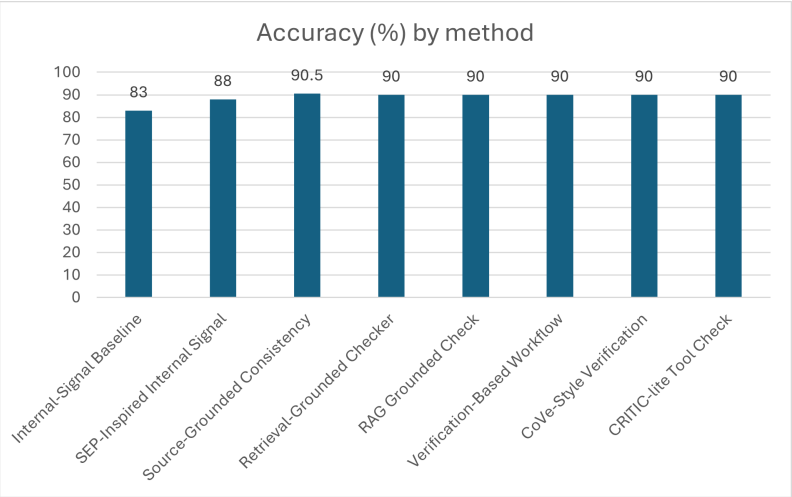

Figure 1: Overall accuracy by method.

Internal Signal showed a similar pattern, although with a better balance, reaching 100.0% precision and 76.0% recall. By contrast, the grounded and verification based methods were more balanced overall, with precision around 89.2% to 89.3% and recall around 91.0% to 92.0%. In the current implementation, this suggests that the evidence based methods provide a stronger practical balance between avoiding false alarms and detecting hallucinated content.

Figure 2 shows that the strongest grounded and verification based methods also created the clearest separation between supported and hallucinated cases. Source-Grounded Consistency, for example, had an average score of 0.179 on supported cases and 0.897 on hallucinated cases. The other grounded and verification based methods followed a very similar pattern, usually falling between 0.19 and 0.23 on supported cases and between 0.87 and 0.89 on hallucinated cases. The internal signal methods also separated supported and hallucinated cases, but the gap was smaller, especially for the baseline model.

4.3 Category-Level Results

Performance also varied across subject areas. Average accuracy across methods was highest in Health at 96.5%, followed by Science at 91.5%, Public Policy at 91.0%, and Education at 90.5%. The weakest areas were Technology at 82.0% and Geography at 85.5%. This shows that some domains were clearly easier than others in the benchmark, although the differences were still moderate enough to allow a meaningful comparison across categories. The category-level averages are summarized in Table 2.

The heatmap (Figure 3) shows two general patterns. First, the relative performance of most methods remains fairly stable across categories. With the exception of the two internal signal methods, most methods stay within a fairly narrow range. Second, differences in domain difficulty seem to affect the internal signal family more strongly than the grounded family. Internal-Signal Baseline dropped to 76.0% in Health and 80.0% in several other categories, while SEP-Inspired Internal Signal ranged

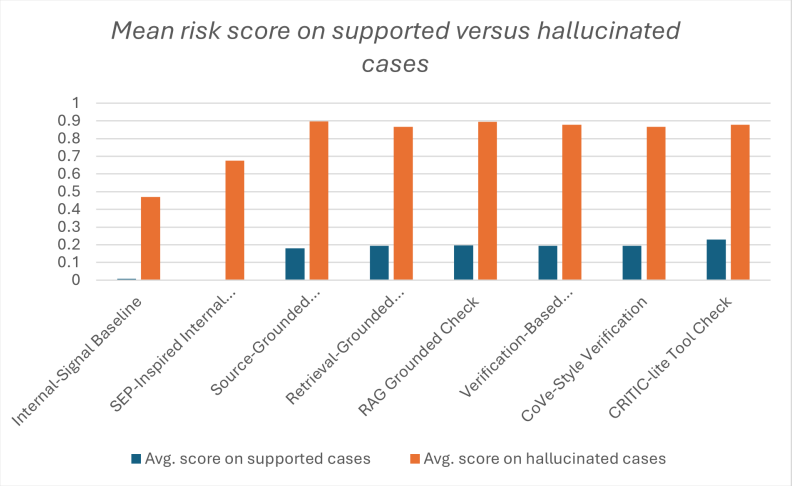

Figure 2: Mean risk score on supported versus hallucinated cases.

Table 2: Average accuracy across methods by category.

Category	Average accuracy across methods (%)
Geography	85.5
Science	91.5
Culture	87.0
Health	96.5
Public Policy	91.0
Environment	87.5
Education	90.5
Technology	82.0

from 72.0% in Technology to 96.0% in Health, Public Policy, and Science. By comparison, the grounded and verification based methods changed less across categories and generally stayed between 84.0% and 100.0%.

4.4 Benchmark Subset Analysis

The benchmark was intentionally divided into direct fact questions and document-grounded memos because these two subsets test different aspects of the prototype. Direct fact questions mainly evaluate whether a method can detect straightforward factual conflicts. Memo cases are harder because they require the use of local evidence, consistency with the source material, and tolerance for short paraphrastic variation. The subset-level results are summarized in Table 3 and illustrated in Figure 5 .

The subset results show one of the clearest differences among the methods in the current system. Direct fact questions were easier for every method than document grounded memos. All grounded and verification-based methods reached 96.9% on direct fact questions. Within that group, Source-Grounded Consistency had the strongest memo performance at 78.6%, while Retrieval-Grounded Checker, RAG Grounded Check, Verification-Based Workflow, CoVe-Style Verification, and CRITIC-lite Tool Check each reached 77.1%. SEP-Inspired Internal Signal reached 92.3% on direct fact questions and 80.0% on memos, while Internal-Signal Baseline reached 87.7% on direct fact questions and 74.3% on memos. Overall, the memo subset was more difficult for every method and highlighted the added challenge of making evidence-based judgments rather than detecting short factual errors.

The supported versus hallucinated split shows a different pattern. Internal-Signal Baseline achieved 99.0% accuracy on supported cases but only 67.0% on hallucinated cases. SEP-Inspired Internal Signal showed a similar pattern, reaching 100.0% on supported cases but only 76.0% on hallucinated cases. By contrast, the grounded and verification-based methods were more balanced across the two sides of the benchmark. Source-Grounded Consistency reached 89.0% on supported cases and 92.0% on hallucinated cases, while Retrieval-Grounded Checker, RAG Grounded Check, Verification-Based

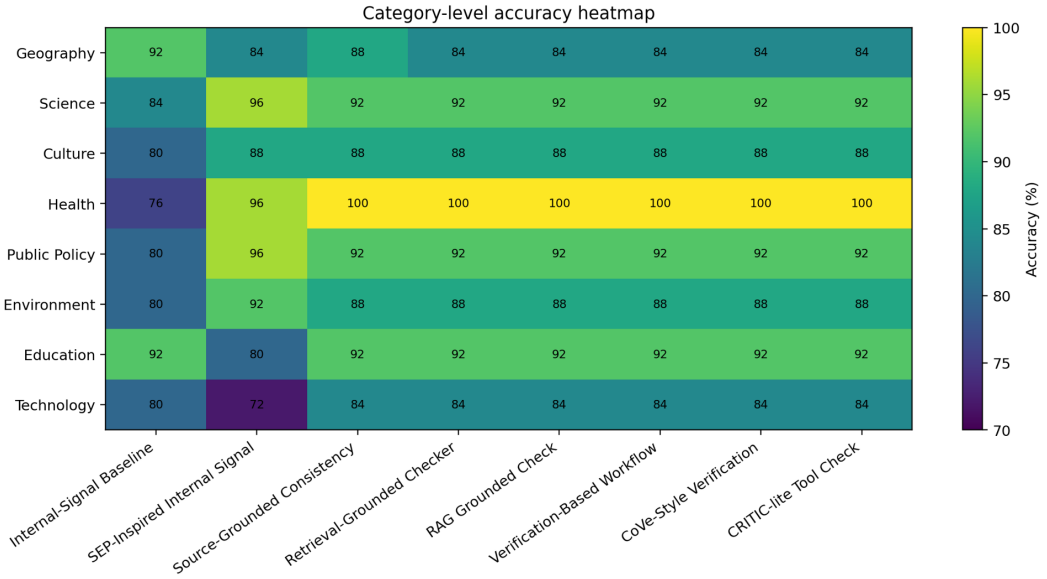

Figure 3: Category-level accuracy heatmap.

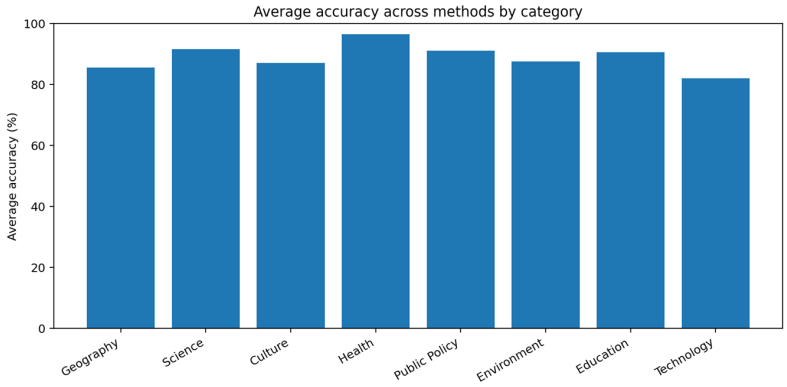

Figure 4: Average accuracy across methods by category.

Workflow, CoVe-Style Verification, and CRITIC-lite Tool Check each reached 89.0% on supported cases and 91.0% on hallucinated cases. This suggests that, in the current implementation, the evidence-based methods provided a more even balance between leaving supported answers unflagged and identifying hallucinated ones.

4.5 Discussion

The results suggest that the detector families behave differently and should not be treated as interchangeable. In the current prototype, the strongest overall results came from the grounded and verification-based methods. Source-Grounded Consistency achieved the highest overall accuracy at 90.5%, while the broader retrieval and verification-based group remained close behind at 90.0%. By comparison, the internal signal methods performed less strongly overall, with SEP-Inspired Internal Signal reaching 88.0% and Internal-Signal Baseline reaching 83.0%. Although the gap is not large, it appears consistently across the benchmark and suggests that, in this prototype, methods with access to explicit evidence were more reliable than methods based only on internal signals.

Source-Grounded Consistency also showed a noteworthy pattern. Even though it uses a simpler checking strategy, it slightly outperformed the retrieval based and staged verification methods. This suggests that added workflow complexity does not always lead to better performance. In this

Table 3: Accuracy on direct fact questions and document-grounded memos.

Method	Direct fact questions (%)	Document-grounded memos (%)
Internal-Signal Baseline	87.7	74.3
SEP-Inspired Internal Signal	92.3	80.0
Source-Grounded Consistency	96.9	78.6
Retrieval-Grounded Checker	96.9	77.1
RAG Grounded Check	96.9	77.1
Verification-Based Workflow	96.9	77.1
CoVe-Style Verification	96.9	77.1
CRITIC-lite Tool Check	96.9	77.1

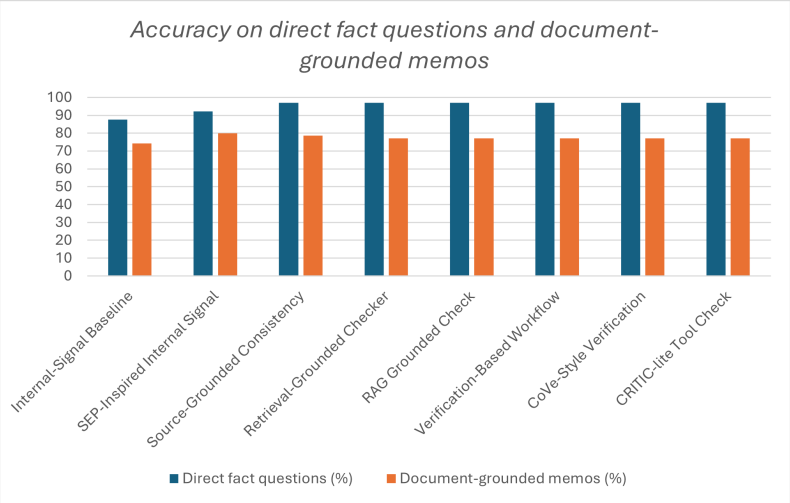

Figure 5: Accuracy on direct fact questions and document-grounded memos.

benchmark, many cases include compact and relevant evidence, which fits well with a direct source grounded checker. Under those conditions, a simpler grounded comparison was often enough to capture most factual conflicts, while the extra retrieval and verification steps offered little quantitative benefit. This also fits the broader SoK perspective, since it shows that the usefulness of a detector depends not only on how complex it is, but also on the evidence conditions in which it is used.

The subset analysis also highlights a clear limitation of the current system. All methods performed better on direct fact questions than on document grounded memos. The evidence-based methods reached 96.9% on direct fact questions, but dropped to 77.1% to 78.6% on memo cases, and the internal signal methods showed a similar decline. This suggests that the prototype is much better at detecting short and explicit factual conflicts than at handling compact judgments that depend on supporting evidence, especially when support has to be inferred across short notes or paraphrased wording. In practical terms, the current system is stronger as a direct factual checker than as a document grounded hallucination detector.

The supported versus hallucinated split shows a clear difference between the detector families. The internal signal methods were highly conservative on supported cases, reaching 99.0% and 100.0% accuracy, but they identified a smaller share of hallucinated answers, with 67.0% and 76.0% accuracy on those cases. The grounded and verification based methods were more balanced, typically reaching about 89.0% on supported cases and 91.0% to 92.0% on hallucinated cases. This balance matters in practice because a useful detector should not only avoid false alarms, but also catch a meaningful share of genuinely problematic outputs. From that perspective, the evidence based methods appear more reliable within the current prototype.

The results should still be interpreted with some caution. The benchmark was intentionally controlled, with short one turn answers, compact evidence passages, and moderate difficulty. It was not designed

Table 4: Accuracy on supported and hallucinated cases.

Method	Supported cases (%)	Hallucinated cases (%)
Internal-Signal Baseline	99.0	67.0
SEP-Inspired Internal Signal	100.0	76.0
Source-Grounded Consistency	89.0	92.0
Retrieval-Grounded Checker	89.0	91.0
RAG Grounded Check	89.0	91.0
Verification-Based Workflow	89.0	91.0
CoVe-Style Verification	89.0	91.0
CRITIC-lite Tool Check	89.0	91.0

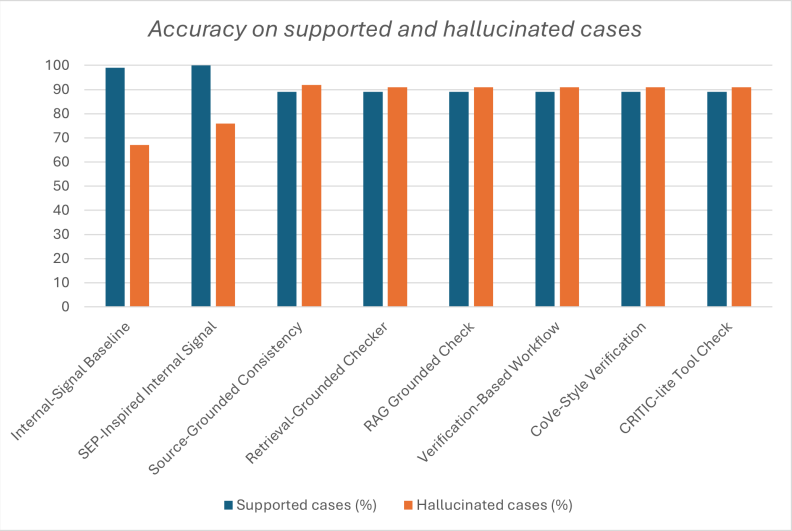

Figure 6: Accuracy on supported and hallucinated cases.

to reflect noisy retrieval, long context reasoning, adversarial paraphrasing, multi hop evidence synthesis, or open web uncertainty. For that reason, the results are best understood as a comparison of detector families within the current prototype under shared conditions, rather than as a broad claim about deployment ready hallucination detection. This does not reduce the value of the project. Instead, it helps keep the contribution clear by showing how different detectors behave under the same setup and by making their assumptions and trade offs easier to see. Overall, the discussion also helps clarify the broader SoK contribution of the project. The experiments do not suggest that one detector family is best in every setting. Instead, they show that different families have different strengths. Grounded and verification based methods perform better when explicit evidence is available, while internal signal methods tend to be more cautious but also less sensitive to hallucinated content. This kind of comparison across detector families is valuable because it goes beyond reporting isolated performance numbers and helps show what different hallucination detectors are actually measuring, when they are most useful, and where their limitations remain.

5 Conclusion

In this report, we examined hallucination detection in large language models from a detector centered perspective. Rather than treating hallucination detection as a single problem with a single best solution, we showed that different detector families rely on different kinds of evidence, make different assumptions, and behave differently in practice. Organizing the field in this way helped us move beyond a broad survey of hallucination and focus more directly on what current detectors actually do, what they require, and where they are most likely to be useful.

The prototype and the benchmark helped make those differences more concrete. In our implementation, grounded and verification-based methods produced the strongest and most balanced overall results, while internal signal methods were more conservative and less effective at identifying hallucinated content. At the same time, the evaluation showed that document grounded memo cases were harder than direct fact questions for every method. This suggests that the current system is better at catching short factual errors than at handling compact judgments that depend on supporting evidence. More broadly, the results do not show that one detector family is always best. They show that performance depends heavily on the evidence available and on the conditions in which a detector is used.

For us, the main value of this project lies in the comparison it provides. The goal was not to present hallucination detection as a solved problem, but to make the differences between detector families easier to understand. Those differences matter in practice because detectors are not interchangeable. Their usefulness depends on what kind of signals they rely on, how transparent their judgments are, and how well they fit the setting in which they are deployed. From that perspective, this SoK provides a clearer and more practical way to compare current approaches by showing where they work well and where their limits remain.

Overall, this project suggests that hallucination detection should be understood as a design space shaped by evidence access, method assumptions, interpretability, and practical cost. A clearer understanding of those factors is important if large language models are to be used responsibly in settings where factual reliability matters. We hope this work helps make that comparison more concrete and also provides a useful starting point for future work on more robust and transferable hallucination detection methods.

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